

KA3

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// pins
const int micPin = A0;
const int ledPins[5] = {2, 3, 4, 5, 6};

// constants
const int N = 800;           // REQUIRED sample size 1000 does not work with
the given arduino
int samples[N];

//Variables
unsigned long startTime;
unsigned long endTime;

float samplingFreq;
float rmsValue;
float signalFreq;

void setup()
{
    Serial.begin(115200);

    // Initialize LEDs
    for (int i = 0; i < 5; i++)
    {
        pinMode(ledPins[i], OUTPUT);
    }
}

void loop()
{
    // ----- DATA ACQUISITION -----
    startTime = micros();

    for (int i = 0; i < N; i++)
    {
        samples[i] = analogRead(micPin);
        delayMicroseconds(50);    // ~20 kHz sampling (stabilizes timing)
    }

    endTime = micros();

    float acquisitionTime = (endTime - startTime) / 1000000.0;
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samplingFreq = N / acquisitionTime;

// ----- SIGNAL PROCESSING -----

// Compute mean (DC offset)
float mean = 0;
for (int i = 0; i < N; i++)
{
    mean += samples[i];
}
mean = mean / N;

// Compute RMS
float sumSquares = 0;
for (int i = 0; i < N; i++)
{
    float centered = samples[i] - mean;
    sumSquares += centered * centered;
}
rmsValue = sqrt(sumSquares / N);

// ----- ZERO-CROSSING FREQUENCY -----
int zeroCrossings = 0;

for (int i = 1; i < N; i++)
{
    float prev = samples[i - 1] - mean;
    float curr = samples[i] - mean;

    // Count only positive-going crossings
    if (prev < 0 && curr >= 0)
    {
        zeroCrossings++;
    }
}

// Correct frequency formula
signalFreq = zeroCrossings / acquisitionTime;

// ----- LED OUTPUT -----
int ledLevel = 0;

if (rmsValue > 5) ledLevel = 1;
if (rmsValue > 10) ledLevel = 2;
if (rmsValue > 15) ledLevel = 3;

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if (rmsValue > 20) ledLevel = 4;
if (rmsValue > 25) ledLevel = 5;

for (int i = 0; i < 5; i++)
{
    if (i < ledLevel)
        digitalWrite(ledPins[i], HIGH);
    else
        digitalWrite(ledPins[i], LOW);
}

// ----- SERIAL OUTPUT -----
Serial.print("Fs: ");
Serial.print(samplingFreq);

Serial.print(" Hz | RMS: ");
Serial.print(rmsValue);

Serial.print(" | Freq: ");
Serial.println(signalFreq);

// ----- SERIAL PLOTTER OUTPUT -----
for (int i = 0; i < N; i++)
{
    //Serial.println(samples[i]);
}
}
```